

Integration of Classy-Blocks and OpenFOAM within Blender

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Weekly Progress Report - Week 3

OpenFOAM GUI

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Introduction

- ◀ **OpenFOAM:** A C++ toolbox for CFD using finite volume methods.
- ◀ **Classy-Blocks:** A Python library for procedural `blockMeshDict` generation, mimicking modern 3D CAD modeling.
- ◀ **Blender Integration:** Utilizing Blender as a frontend for geometry extraction and mesh visualization using `pyvista`.

Research Gap

- ◀ **Gap 1:** Lack of native Blender 5.x support for advanced Python meshing libraries like `classy_blocks`.
- ◀ **Gap 2:** Absence of an integrated “One-Click” solution for geometry extraction, mesh building, and quality reporting within a unified environment.

Problem Statement

- ◀ To develop a Blender Add-on that leverages `classy_blocks` and `pyvista` to automate OpenFOAM case setup, allowing users to extract geometry, generate high-quality meshes, and execute solvers without leaving the Blender interface.

Work Done: Stabilization & Quality Metrics

- ▶ **Operator Safety:** Added mandatory directory existence checks (`os.path.isdir`) preventing operator crashes when `case_path` is empty or invalid.
- ▶ **Actionable Feedback:** Replaced vague Python exceptions with context-aware `self.report()` calls guiding the user to specific UI fixes.
- ▶ **Quality Parsing:** `parse_mesh_quality()` captures combined `stdout/stderr` to accurately extract non-orthogonality and skewness metrics from the solver output.
- ▶ **Metrics UI:** Automatically stores the latest solver metrics in a scene custom property and visually displays them within the 3D Viewport sidebar.

Work Done: Meshing Configuration UI

- ▶ **Grading Controls:** Introduced `grading_type` (Ratio, Start Size, Symmetric). `mesh_builder.py` dynamically dispatches `_apply_chops()`, leveraging `classy_blocks` auto-computation.
- ▶ **STL Projection:** Implemented `stl_file` and `stl_projection_face` properties to map block faces to surface triangulations for complex geometry projection.
- ▶ **Auto-Update Ready:** Added `use_auto_update`, `limit`, and `delay` properties to the UI to configure future automated background meshing pipelines.
- ▶ **Verified Stability:** All feature interactions tested. Added 3 new grading tests generating accurate `blockMeshDict` files. Pipeline now fully stable at 32/32 tests passing.